

Lecture

Ethics and Philosophy of Mind
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I'm not a philosopher,
I'm a Associate Prof. of AI
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The Rulebook for Today

- There are no “correct” answers
- But there’s more than just “opinions”
- We’re looking for well argued positions
- I’ll be clear about my own positions
- I’ll often ask you for yours
- My goal: get you to think and discuss (and maybe read)

This is **not** the definitive lecture on
Ethics or Philosophy of Mind,
just an intro of ideas and questions

I'm not a philosopher, and not an ethicist

I'm a Associate Prof. of AI
Knowledge in Artificial Intelligence
Computer Science Dept. VU

The goal of today is to make you reflect
on who you are and what you do.

Overview

1 Ethics

2 Philosophy of Mind

Part 1: Ethics

Ethics are moral principles that guide our behavior.

Today's ethics: questions we face now

Sci-Fi ethics: questions we face down the road

Some example questions:

1. How do we encode ethical behavior?
 - a. Should we construct ethical rules for intelligent agents (ethics in design)?
 - b. Should we build intelligent agents that reason over ethics (ethics by design)?
2. (How should we behave towards AI?)
3. How does the existence of AI affect our daily lives?

Part 1.1: Some thoughts on Ethics

Morality vs Law

- Not all unethical behaviour is illegal

Cheating in a football game (time wasting)

- Not all illegal behaviour is unethical

Crossing at a red light when there is no one remotely close.

- Not all legal behaviour is ethical

Spreading horrible rumors about other people.

Two Types of Ethics

Radical Freedom
(ethics by design)

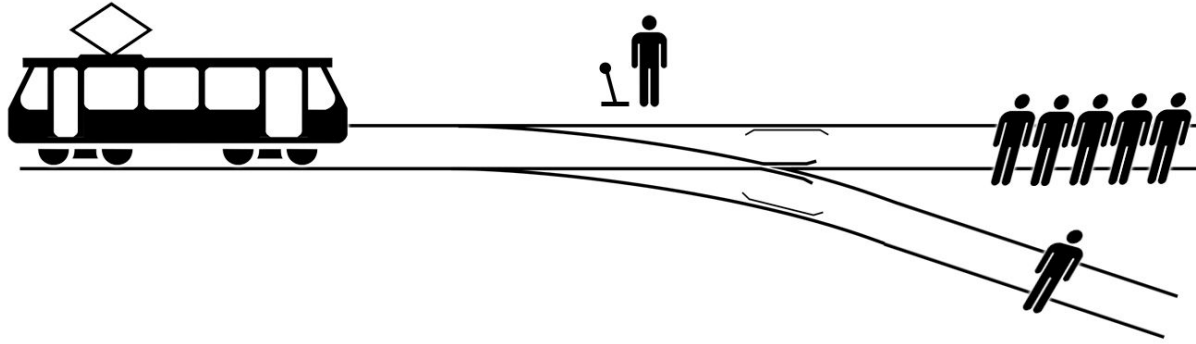
Moral Absolutism
(ethics in design)



1. A robot may not injure a human being or, through inaction, allow a human being to come to harm.
2. A robot must obey the orders given it by human beings except where such orders would conflict with the First Law.
3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Laws.

Part 1.2: Why is Ethics important? (or some Ethical Dilemmas)

Trolley Problem

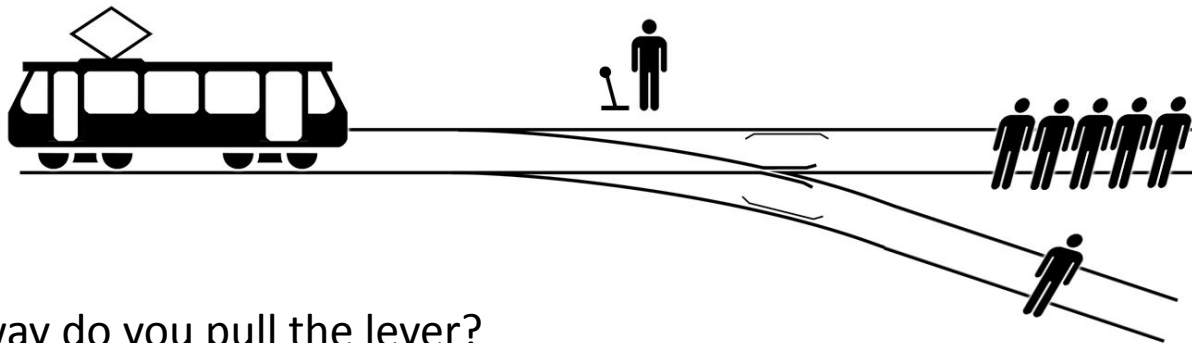


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Trolley Problem



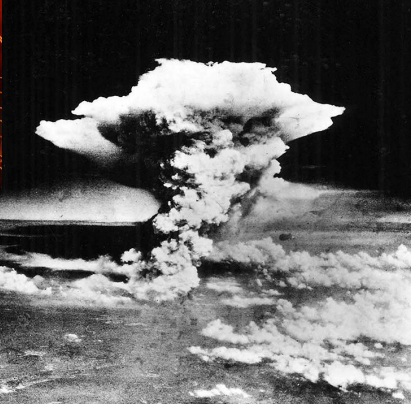
Which way do you pull the lever?

Fat man variation: either you push a fat man on the tracks or allow 5 people further down the tracks to die.

How would a self-driving car choose when faced with two options?

Think with a [veil of ignorance](#), ignoring your personal biases and situation.

Manhattan Project

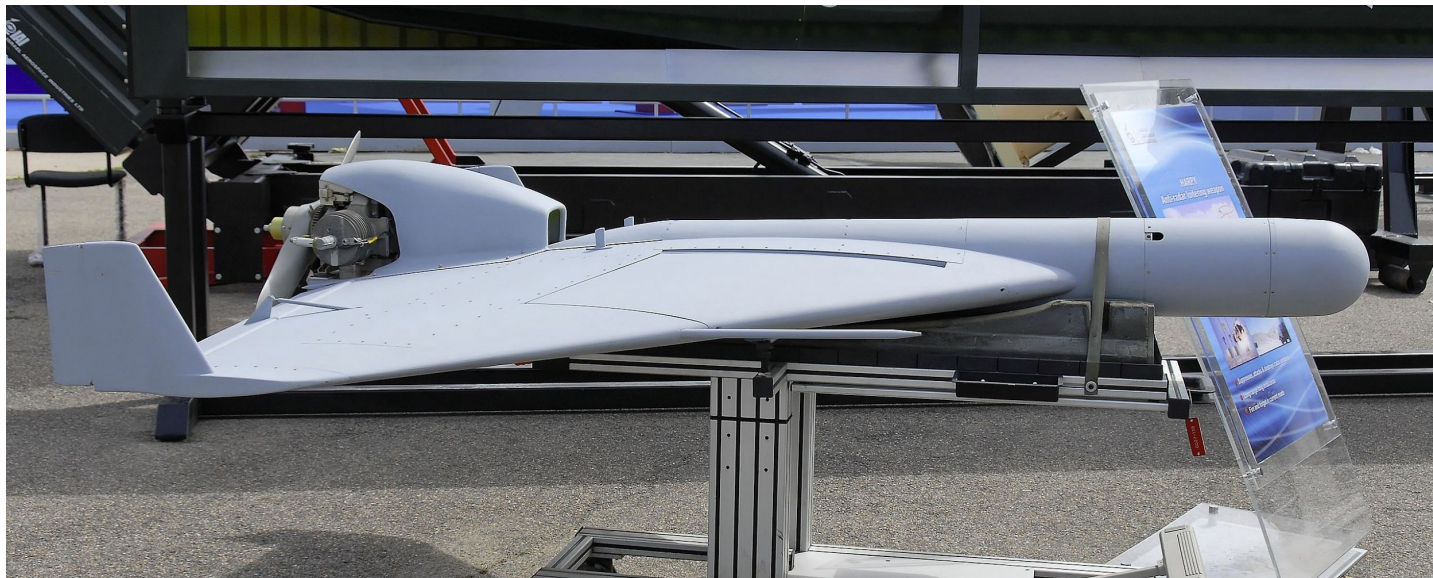


MENTIMETER

Part 1.3: Some of Today's AI Ethics Problems

- Autonomous Weapons
- Privacy
- Value-based Design
- Bias/ Profiling

Autonomous Weapons



IAI Harpy
Israeli missile for targeting radar systems.

Human-in-the-loop

Human-in-the-loop systems require human intervention before they can act.

Proposal from the Dutch government on this topic (translated):

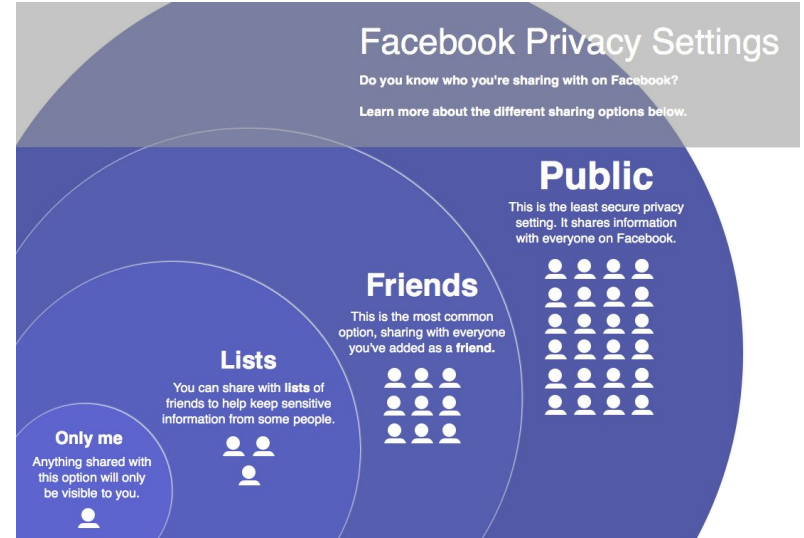
“The AIV/CAVV is of the opinion that the the loop in the broader sense, the **wider loop**, should be abided by. Prior to the weapon’s decision to select a target for attack, a human has made the decision to deploy the weapon and the weapon has been programmed, whereby decisions have been made for the selection of a target.”

Privacy



Stasi Surveillance

We accept that state surveillance, like this example from former east Germany, is bad.



Company Surveillance

But give away our data to other entities freely.

Note: not only ethics but also societal questions....

The previous observation points to very serious political and societal challenges and questions:

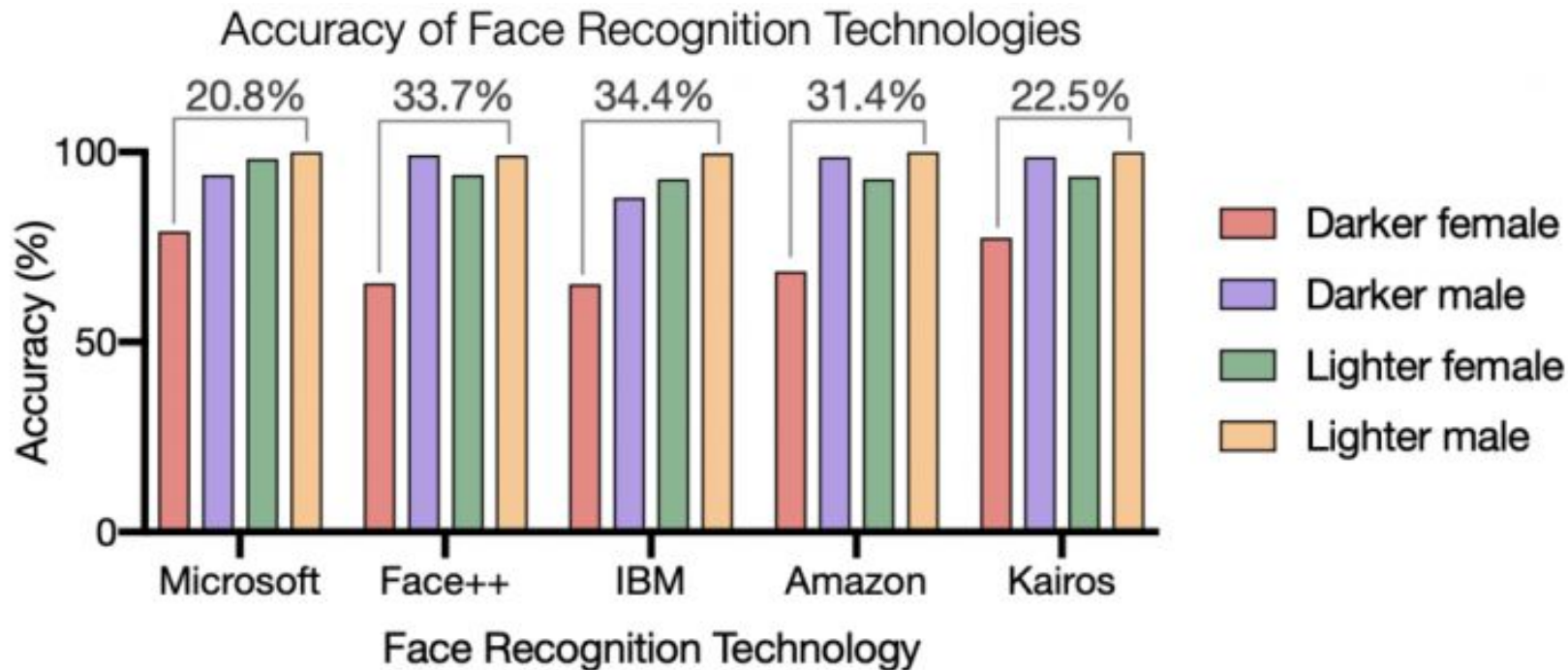
- Who controls data?
 - Services require huge amounts of data
- Who controls knowledge?
 - Wikipedia is highly biased towards rich and powerful (and thus everything that is trained on it)
- Who controls technology?
 - Think of satellite technology in Ukraine
- Who owns infrastructure?
 - LLMs require extreme levels of infrastructure
- Is knowledge, data, technology and infrastructure distributed in fair ways?

... this is out of the scope of this presentation.

Value-free design (historical example)



Value-free design (in Artificial Intelligence)



Profiling

Profiling is using someone's data to evaluate personal aspects of the individual.

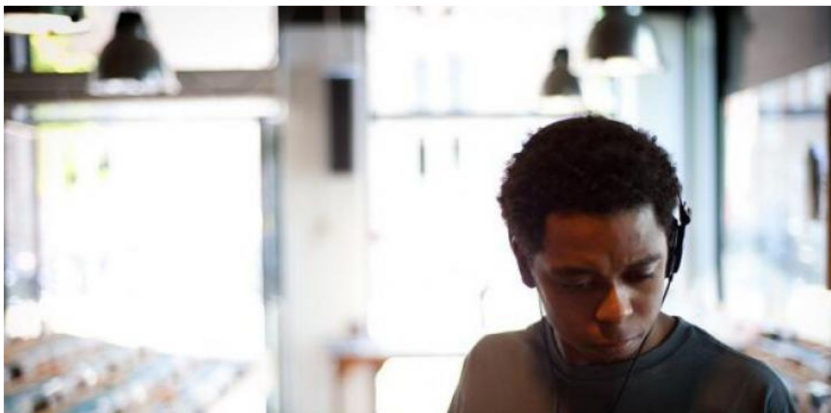
We do this in real life...

But AI does it too...

TOP STORIES

POLICE RACIAL PROFILING OVERWHELMINGLY APPROVED BY DUTCH PUBLIC

By Janene Pieters on June 6, 2016 - 09:02



The Washington Post

Democracy Dies in Darkness

INNOVATIONS

These robots were trained on AI. They became racist and sexist.

As billions flow into robotics, researchers who conducted the study are concerned about the effects this might have on society

As part of a recent experiment, scientists asked specially programmed robots to scan blocks with people's faces on them, then put the "criminal" in a box. The robots repeatedly chose a block with a Black man's face.

Prosecutor's Fallacy

Prosecutor's fallacy states that 1) if the accused were guilty, the probability of evidence is high and thus 2) given the evidence at hand, the probability that the accused is guilty is high.

Abuse of conditional probability

$$p(\text{black}|\text{criminal}) \text{ vs. } p(\text{criminal}|\text{black})$$

Racial profiling is a prosecutor's fallacy.

Is this so bad? Yes. This leads to increased minor misdemeanor detection.

Why do we not just un-bias the data?

Idea: remove “race” or “gender” as a feature before deploying AI for decision making?

Problem: the feature “race” is not the only indicator of race.

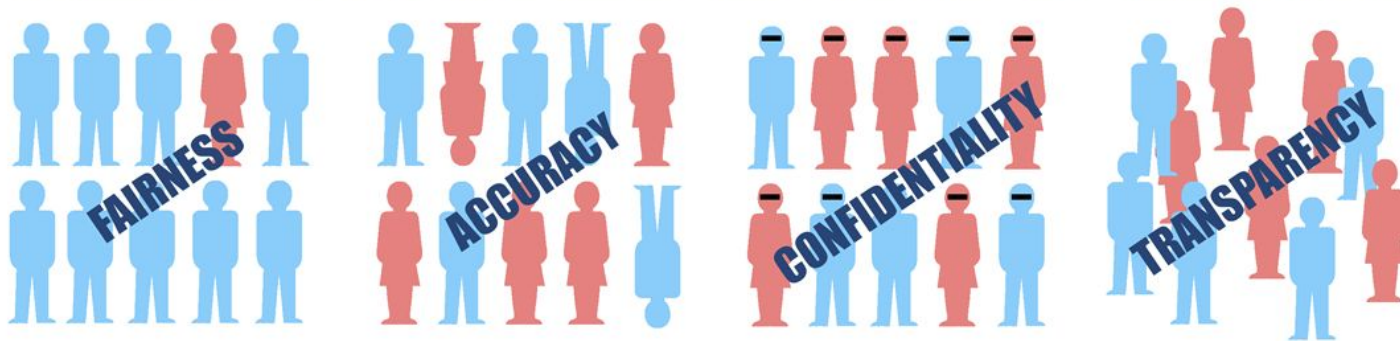
- Postal code
- Profession
- Average salary
- Mode of transport
- Etc.

Then, how do we disallow the use of sensitive characteristics?

Proxies for Gender

Hours worked per week
Military service
Gave birth to a child last 12 months
Educational attainment
Marital status
Ever used Veterans Health Administration
Science and Engineering degree flag
Recoded field of degree
Time of arrival at work
Hearing difficulty

RESPONSIBLE DATA SCIENCE



Part 2: Philosophy of Mind (slides from Frank van Harmelen)

Traveling from small to large

1. Quick poll on some classical questions
2. What is Intelligence
3. The Mind-Body problem
4. Can a computer be intelligent?
5. Symbol grounding
6. Intentional stance, epiphenomenalism
7. Free will

In AI Philosophical Questions are never far away

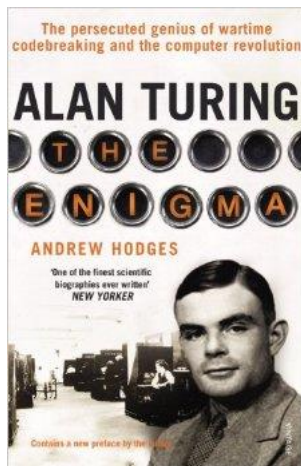
- Can computers be intelligent?
- Can computers have consciousness?
- Can a computer be creative?
- Can a computer have emotions?
- Can computers have free will?

Part 2.1: Intelligence

What is Intelligence?

Classically people tried to define Machine Intelligence via the Turing Test

The objective is to compare natural (human) versus artificial intelligence



Difference between human and machine intelligence



Intelligence = everything a computer cannot do?

Some examples of the horizon effect:
(movies on the next slides)

- Chess?
- Autonomous Navigation?
- Unlimited Question Answering?



Chess: we thought one would have to be intelligent



Robots: we thought they would have to be intelligent



Part 2.2: Can Computers ever be Intelligent?

The Mind-Body problem

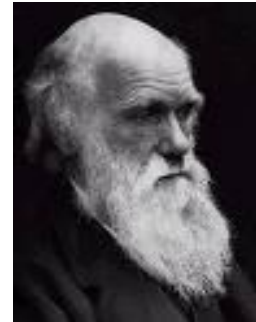
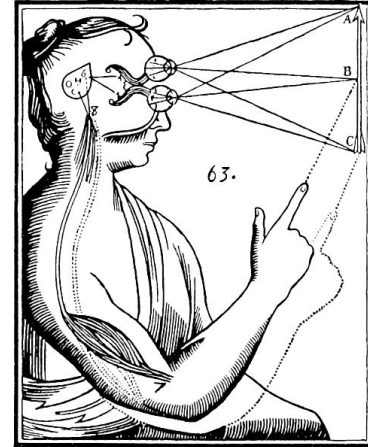
Descartes



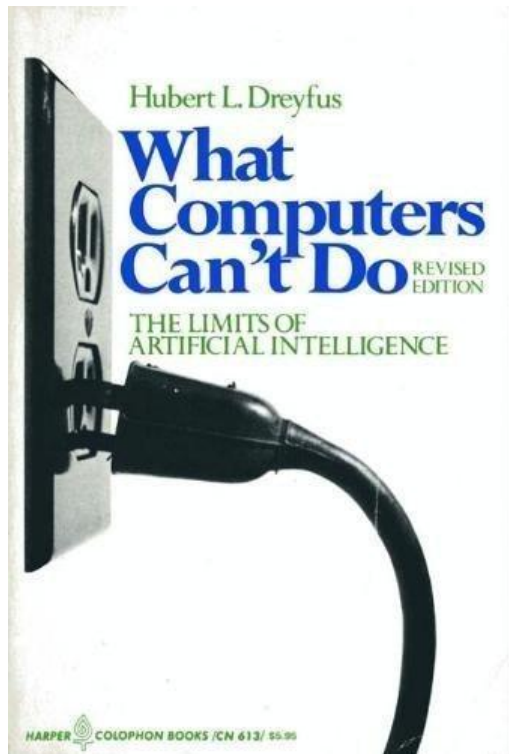
Interactionism



Materialism



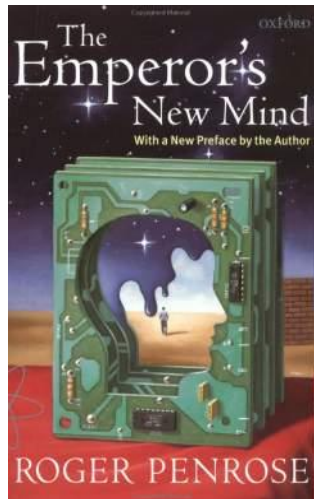
What Computers Can't Do



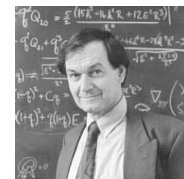
A thought provoking book from the 1970's on the inherent limitations of computer intelligence (based on symbol manipulation).

A great misunderstanding accounts for public confusion about thinking machines, a misunderstanding perpetrated by the [unrealistic claims researchers in AI](#) have been making, claims that thinking machines are already here, or at any rate, just around the corner.

Can a Computer be Intelligent?



- Substitution argument
(one neuron at a time)
- Medium argument
(what's so special about carbohydrates?)
- Chinese Room argument
(Symbol Grounding)
- Formal systems argument



Roger Penrose

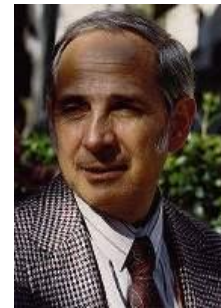
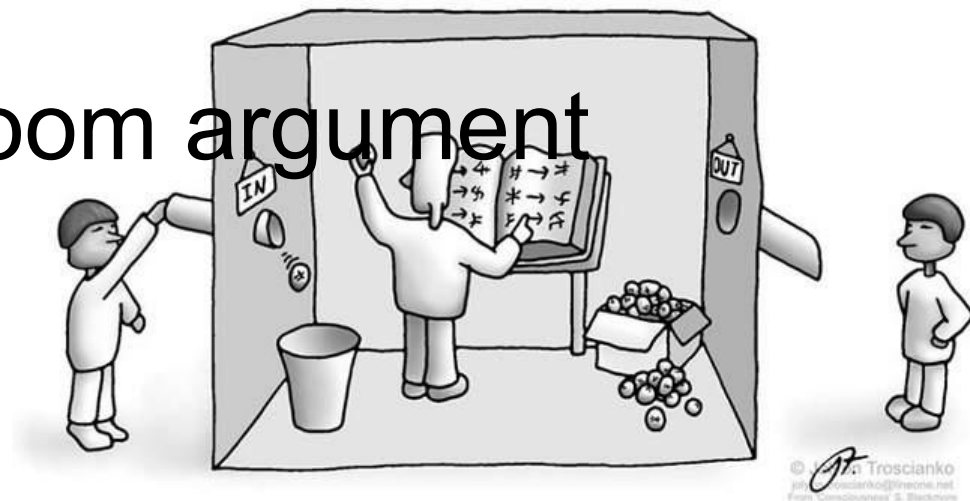


John Searle



Gödel

Chinese Room argument

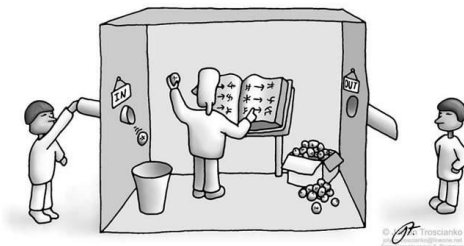


Does anybody/anything here understand Chinese?

Chinese Room argument

An important objection to this thought experiment is that it could never be done in reality, because such a rule-book could never exist.

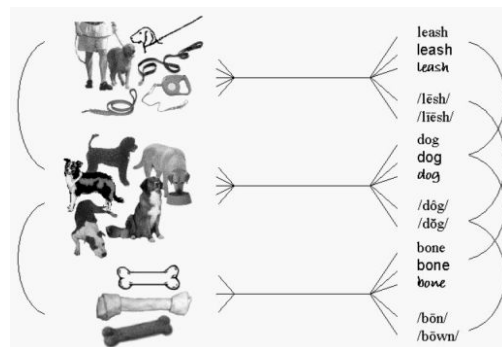
Understanding the questions (in Chinese or otherwise) will always require understanding the “real” world, through sensory perception: how could you ever understand the symbol “red” if you have never seen it?



彤

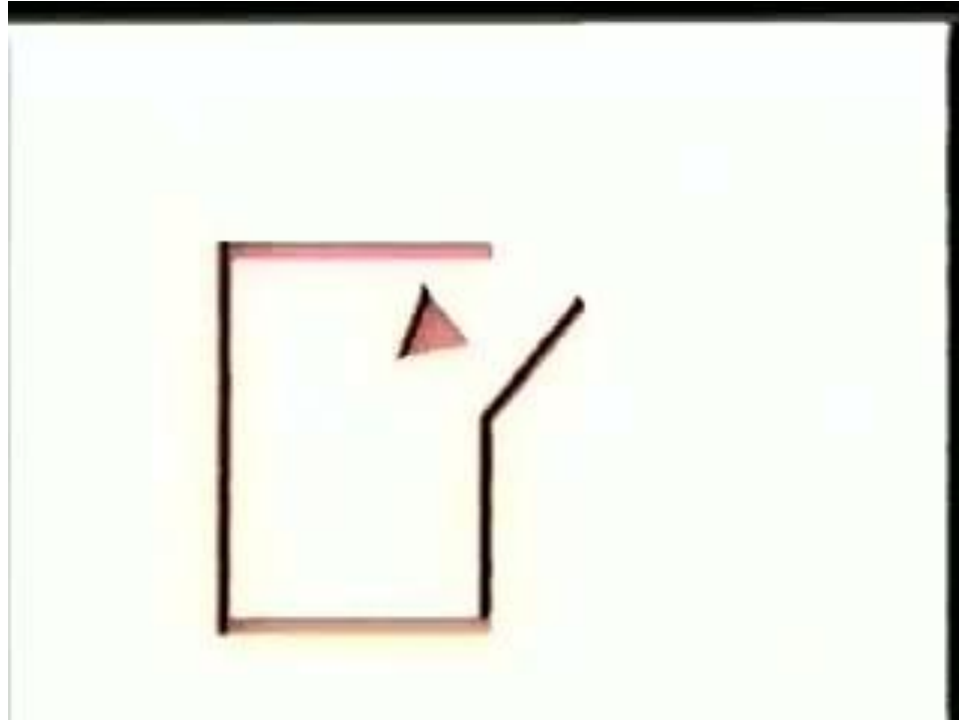
Symbol Grounding 2/2

- How can a computer ever know anything about the “real” world?
- Symbols can only refer to other symbols
- How can a computer ever know what is “red”, or “heavy” or “sad”, or ...etc.
- Therefore,
simulated intelligence $\neq$ real intelligence!



Part 2.3: Consciousness

Give live commentary on this video



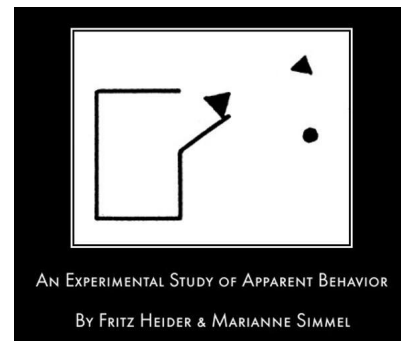
Intentional Stance

What did you think when you watched this?

http://www.youtube.com/watch?v=wp8ebj_yRI4

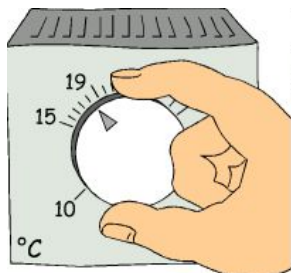
What proof do I have

- for the “consciousness” of other people?
- For their motivation?
- For their emotions?



Intentional Stance (Daniel Dennett)

- Intelligence/consciousness is “attributed”
- Intelligence/consciousness is “gradual”
- Attribute as much as is useful



Intentional Stance

- If its **useful** to talk about consciousness, motivation, feeling, etc, then we are allowed to do so (or even: should do so), equally for people and machines.
- People have a strong tendency to take the **intentional stance**:
 - animals (even very primitive ones)
 - machines (even if we know how they work)
- And **therefore** we will *call* our computers “intelligent” and “conscious”



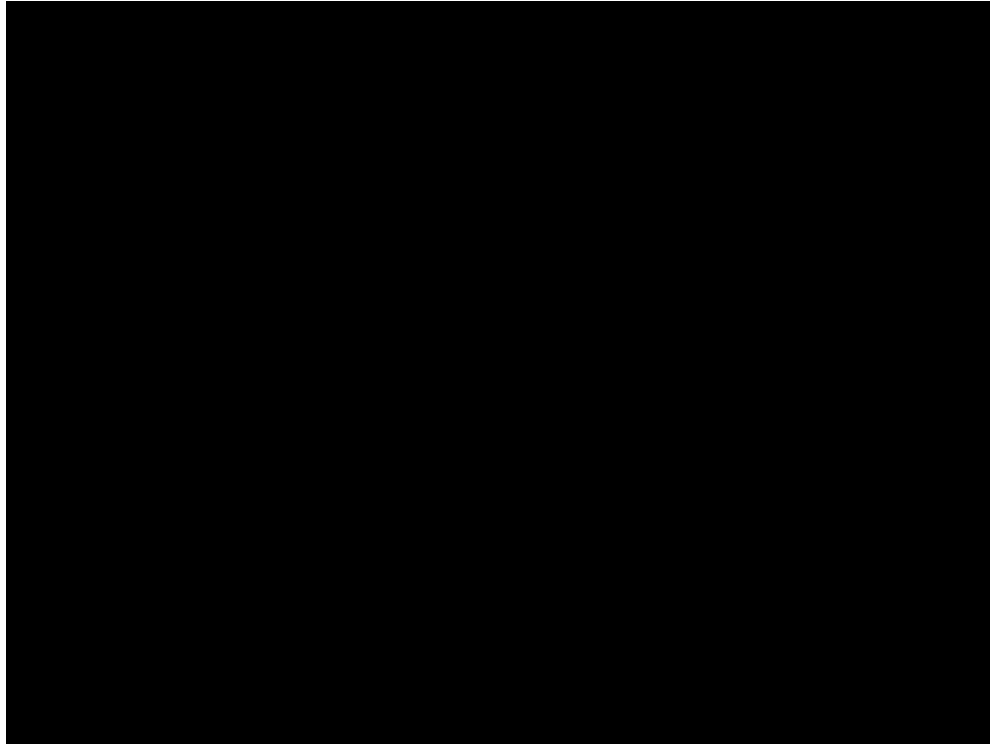
AI – the Movie

An illustration on how easily people adopt the intentional stance



Her – the Movie

Another illustration how easily people adopt the intentional stance



What about “self-consciousness” that we all feel?

(Maybe) self-consciousness is just the story we tell

- about ourselves
- to ourselves

in our attempt to maintain between the world and ourselves

Technically:

we're applying the intentional stance to ourselves!

Consciousness = ?



or



Part 2.4: Free Will

Questions on Free Will and Agency

- do we humans have free will?
- do monkeys have free will?
- do dogs have free will?
- do ants have free will?
- could computers ever have free will?

Is Hybrid Intelligence possible?

Hybrid Intelligence is the study of Artificial Intelligent systems that **cooperate synergistically**, proactively and purposefully with humans, amplifying instead of replacing human intelligence.

Ben Shneiderman (HCI): While the goal of “amplifying instead of replacing human intelligence” is central to my thinking of what defines HCAI, I hold an extreme view, preferring to ensure human control of supertools and tele-bots rather than having AI systems “cooperate” with people or participate in “hybrid” systems. For me, people are a different category from machines, because humans are legally and morally responsible for uses of technology. I encourage participation in this conference, but hope the organizers and attendees will clarify their position and strengthen the theme of amplifying human intelligence and ensuring human control. It may be too much to ask, but I hope the conference and their center name might be changed to something like “Amplifying Human Intelligence with AI.”

Can there really be true Hybrid Intelligence?

P1: AI can perform certain intelligent actions (from experience).

P2: AI is capable of actions (based on P1).

P3: To be capable of performing actions implies agency (from the definition of agency).

P4: AIs could be said to have **agency** (from P2 and P3).

P5: Not all actions are **free actions** (even in the human person, from experience).

P6: It takes not only rationality but, most importantly free-will to perform free actions (even in the human person).

P7: AI does not have free-will (if one does not hold a materialistic-deterministic non-causal view of the person and action).

P8: AI cannot perform free actions (based on P6 and P7).

P9: AI is not a free agent (from P3, P6, P7, and P8).

P10: Free agency implies autonomy (from the definition).

P11: AI cannot have autonomy (from P9).

Φιλοσοφία
Volume 24, 1: 2023

ACTION AND AGENCY IN ARTIFICIAL INTELLIGENCE: A PHILOSOPHICAL CRITIQUE

Justin Nnaemeka Onyeukaziri
Fu Jen Catholic University, Taiwan

The objective of this work is to explore the notion of "action" and "agency" in artificial intelligence (AI). It employs a metaphysical notion of action and agency as an epistemological tool in the critique of the notion of "action" and "agency" in artificial intelligence. Hence, both a metaphysical and cognitive analysis is employed in the investigation of the quiddity and nature of action and agency per se, and how they are, by extension, employed in the language and science of artificial intelligence. The advent of the science of artificial intelligence and cognitive science, and the technological applications of artificial intelligence in the production of agents such as driverless cars and expert systems, have raised the question of moral, ethical and/or legal responsibility in AI agents. This has re-emphasized the importance of the philosophical discourse on the notions of action and agency, which in contemporary intellectual discourse are now perceived to be phenomena within the epistemic competence of the natural sciences. This paper argues that AI systems do not and cannot possess free agency and autonomy; thus, cannot be morally and ethically responsible. Hence, it recommends a socio-political response to the question of responsibility in AI. It is then the duty of individual nations, or the global community to define and enact policies on who shoulders the responsibility of actions executed by AIs.

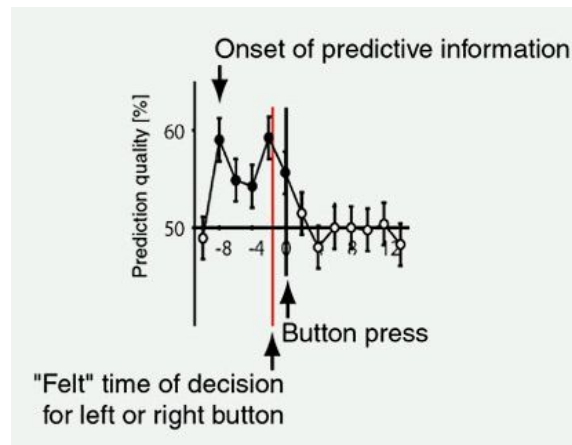
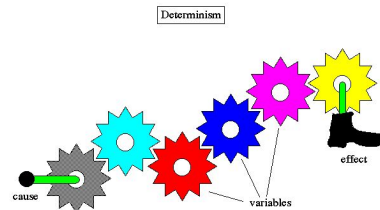
Keywords: Action, agency, Artificial Intelligence, person, mind, consciousness.

INTRODUCTION

With the advancement in the development of Artificial Intelligence (AI), both as the science of intelligence, and as technology that simulates human intelligence in decision-making and problem-solving, the questions on the moral, ethical,¹ and legal responsibilities of AI actions are becoming more frequent and prominent. These questions are definitely outside the epistemic framework and authority of the science

Free Will

- It violates Physics because the physical world is deterministic, and leaves no room for “free” will
- It violates Neuroscience



A Free Will worth wanting

Daniel Dennet again: We must **give up the “naïve”** version of free will that we all grow up with

But: there other views on free will which still rescue the crucial idea of “responsibility for your actions”

These other views rely on the idea that we can **“represent”** our reasons for acting, and that we can reflect on them, and that **this reflection can lead to change.**

A Free Will that is possible but still worthwhile (2'15, 3'50)



Finally

- These are some of the deepest questions of the 21st century
- They touch deeply on our self-image & on our morality,
- We should have these debates now
- We should not leave those discussions to the philosophers.

